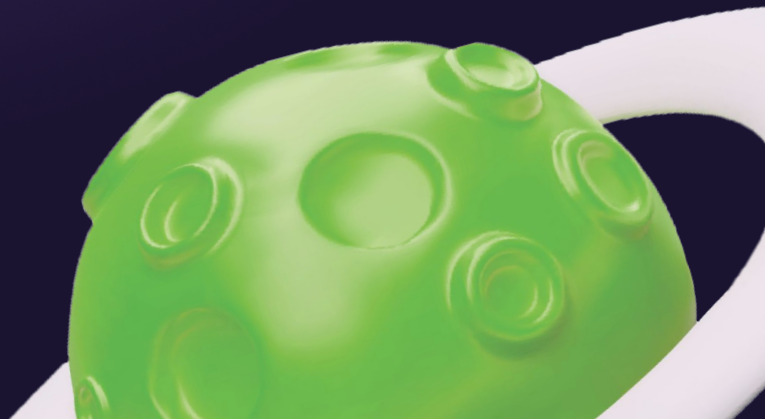
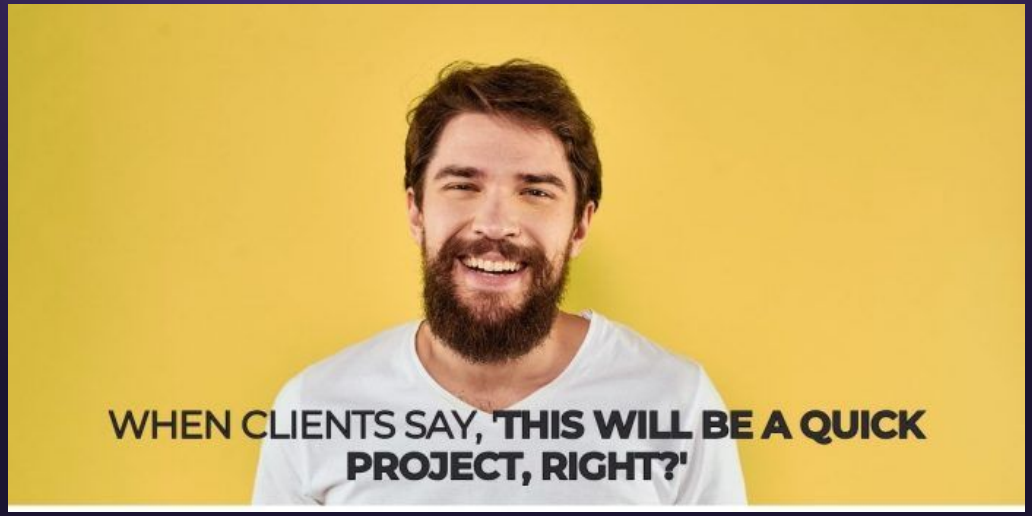
Black Hole Team

We are not going to get pulled!

Meeting - April 16th




WHEN CLIENTS SAY, 'THIS WILL BE A QUICK PROJECT, RIGHT?'

We are building **ONE thing only**:

👉 A **web platform where users interact with AI agents**

- AI agent = digital assistant
- Connects the user to sub-agents depending on the task the user needs to complete.

What it does:

- Understands the request
- Decides **HOW to solve it**
- Chooses which agents to use
- Combines results
- Returns final answer

👉 It does NOT do everything itself

👉 It **delegates**

Example

User says:

"Analyze market trends and give me a report"

Manager Agent receives request

"I need research + analysis + formatting"

MA calls from User's personal pool of agents:

Research Agent → Analysis Agent → Report Agent

Collects results

Sends final answer to user

WHERE DO THIS AGENTS COME FROM?

👉 **Sub-agents = tools in a toolbox**
👉 **Sub-agents are CREATED by users (or the team) and stored in the system.**
👉 They are **defined, saved (in the DB Agent entity/table), and reused**.
👉 They become part of the **Global Source Pool** → “Manager Agent only sees **personal** pool”

📌 “Each agent is configured with system prompt + tools + skills”

```
Name: Web Research Agent

System prompt:
"You are an expert researcher. Search the web and summarize results clearly."

Tools:
- Web search
- HTTP requests
```

THE FLOW MAPPED TO THE BRIEFING

01

User → Frontend

User types message in UI

>>> REACT (frontend)

User → UI → API

02

Frontend → Backend

Frontend sends:

POST /conversations/{id}/messages

👉 Backend receives request (Spring Boot API)

03

Backend → Manager Agent

THE FLOW MAPPED TO THE BRIEFING

04

MA receives:

1. User's request message
2. List of agents

THINKS which ones is going to uso for the task

👉 This decision comes from the LLM

05

MA chains / delegates sub-agents:

- Manager → Research Agent
- Manager → Analysis Agent
- Manager → Excel Agent

06

Subagent executes task

Example: Research Agent:

- Call tool → Web search
- Get results
- Send results to LLM
- Produce output

Agent → Tool → Result
→ LLM → Output

THE FLOW MAPPED TO THE BRIEFING

07

Backend stores execution.

- WorkflowExecution created
- Step results stored
- Token usage tracked

👉 This is also a **tracking + analytics system**

08

Workflow chaining:

“output of one agent becomes input of next”

09

Manager Agent builds FINAL answer:

- collects outputs

- synthesizes response

THE FLOW MAPPED TO THE BRIEFING

10

Backend → Frontend

Response is streamed: WebSocket → UI

“streamed tokens via WebSocket”

11

User sees result”

👉 Final output appears live

12

Manager Agent builds FINAL answer:

- collects outputs
- synthesizes response

FINAL TEAM STRUCTURE (5 PEOPLE)

Person	Role	Responsibility	Notes
P1	Backend Core + DevOps owner	Auth, Users, Security, Base project, Docker	🔥 Most critical
P2	Agent System	Agents + execution	Core product
P3	Workflow Engine	Automation logic	Depends on Agents
P4	Tools + MCP	Plugins, tools system	Parallel work possible
P5	Frontend + Conversations	UI + Chat + Manager agent	Connects everything

MEET THE TEAM

P1: Backend Core (Foundation)

```
[ Backend Core ]
   ├── Authentication (login/register)
   ├── Users & Roles
   ├── Security (JWT)
   └── Project base (API + DB)
```

👉 Builds the foundation of the system

“This role ensures users exist, are authenticated, and that the system is secure and structured.”

P2: Agent System (Core Logic)

```
[ Agents ]
   ├── Create agent
   ├── Configure (prompt, skills)
   └── Execute (AI call)
```

👉 Builds the brain of the system

“Agents are the core intelligence — they receive instructions and generate responses.”

MEET THE TEAM

P3: Workflow Engine (Automation)

```
[ Workflows ]
   ├── Step 1 → Agent A
   ├── Step 2 → Agent B
   └── Step 3 → Output
```

👉 Automates sequences of actions

“Workflows allow chaining multiple agent actions to automate complex tasks.”

P4: Tools & Plugins (Extensions)

```
[ Tools ]
   ├── API calls
   ├── External data
   └── Plugins (MCP)
```

👉 Extends what agents can do

“Tools allow agents to interact with the outside world, like APIs or databases.”

MEET THE TEAM

P5: Frontend + Manager (User Layer)

👉 Connects user with the system

“This layer handles user interaction and decides how to route requests inside the system.”

```
[ Frontend ]
   ├── Chat UI
   ├── Dashboard
   └── User interaction

[ Manager Agent ]
   └── Chooses:
        → which agent
        → which workflow
```

CRITICAL DEPENDENCY MAP

```
Auth (P1)
    ↓
Agents (P2)
    ↓
Workflows (P3)
    ↓
Manager / Conversations (P5)

Tools (P4) → connects to Agents (parallel)
Frontend (P5) → connects to EVERYTHING
```

👉 If you break this order → **team gets blocked**

GLOBAL EXECUTION PLAN

🔹 PHASE 0 — SETUP (Day 1–2) ⚠️ ALL TOGETHER

Task	Who
Docker + Postgres + backend empty	P1
`/health` endpoint	P1
Basic frontend (empty React app)	P5
Connect frontend → backend	P1 + P5

👉 MUST WORK: docker compose up

🔹 PHASE 1 — FOUNDATION (Week 1–2)

Task	Person	Dependency
Auth (login/register + JWT)	P1	NONE
User roles (EMPLOYEE/MANAGER)	P1	NONE
Agent CRUD (no AI yet)	P2	needs User
Basic UI (login + dashboard)	P5	parallel

👉 At the end:

✔ Users exist
✔ Agents can be created
✔ UI can call API

🔹 PHASE 2 — CORE PRODUCT (Week 2–3)

Task	Person	Dependency
Agent execution (basic AI call)	P2	Agent CRUD
Conversations (chat system)	P5	Auth
WebSocket (chat streaming)	P5	Conversations
Tool system base	P4	parallel

👉 At the end:

✔ You have a **ChatGPT-like system**

🔹 PHASE 3 — AUTOMATION (Week 3–4)

Task	Person	Dependency
Workflow engine	P3	Agents
Workflow execution	P3	Engine
Tool integration with agents	P4	Tools + Agents
UI for chat + agents	P5	previous

👉 Now your system becomes **useful**

🔹 PHASE 4 — INTELLIGENCE LAYER (Week 4–5)

Task	Person	Dependency
Manager Agent (decision making)	P5	Agents + Workflows
Agent orchestration	P5	Manager
Advanced tools/plugins	P4	Tools

👉 This is the **“wow factor”**

🔹 PHASE 4 — INTELLIGENCE LAYER (Week 4–5)

Task	Person	Dependency
Manager Agent (decision making)	P5	Agents + Workflows
Agent orchestration	P5	Manager
Advanced tools/plugins	P4	Tools

👉 This is the **“wow factor”**

🔹 PHASE 5 — POLISH + MODULES (Final weeks)

Task	Who
Security improvements	P1
UI polish	P5
Analytics / extras	Team
Testing + fixes	ALL

THANK YOU